

Tree-width and Path-width of Comparability Graphs of Interval Orders

Renate Garbe

Department of Applied Mathematics, University of Twente, P.O.Box 217
NL-7500 AE Enschede, The Netherlands, garbe@math.utwente.nl

Abstract. The problem to decide whether the tree-width of a comparability graph is less than k is NP -complete, if k is part of the input. We prove that the tree-width of comparability graphs of interval orders can be determined in linear time and that it equals the path-width of the graph. Our proof is constructive, i.e., we give an explicit path decomposition of the graph.

1 Introduction

The *tree-width* of a graph can be defined as the smallest number k such that the graph can be embedded into a k -tree, where a k -tree is a graph that can be reduced to the complete graph on k vertices by a sequence of removals of vertices of degree k whose neighborhood is completely connected. Graphs with tree-width k are also known as *partial k -trees*.

Equivalently to the above definition the tree-width of a graph G can be defined to be the smallest number k such that G can be embedded into a *chordal graph* with *clique number* k (cf. [ACP 87]). If the chordal graph is required to be an *interval graph* we get the definition of the *path-width* of a graph G . Because in general the path-width of a graph exceeds its tree-width an interesting problem is to investigate whether there are classes of graphs whose path-width and tree-width are equal.

The notions of tree-width and path-width were introduced by Robertson and Seymour [RoSe 83] and [RoSe 86] in the framework of graph minor theory. They showed by non-constructive methods that for fixed k the problem of determining whether a graph has tree-width less than k is polynomially solvable.

Their result was improved by Bodlaender and Kloks [BoKl 91] who gave an algorithm of order $O(n \log^2 n)$ for each constant k to determine whether the tree-width or path-width of a given graph is less than k . The algorithm, however, depends exponentially on k .

Arnborg, Corneil and Proskurowski [ACP 87] proved that if k is part of the problem instance then the problems to decide whether a graph G has tree-width or path-width less than k is NP -complete.

Graphs with bounded tree-width or path-width are of special interest because many problems on graphs that are intractable in general can be solved by polynomial algorithms when restricted to graphs with bounded tree-width or path-width.

See for example the *List Coloring* Problem, a generalization of the problem of determining the *chromatic number* of a graph, given by Jansen and Scheffler [JS 1992]. More examples and a survey on tree-width and path-width are given in Proskurowski and Sysło [PrS 90] and Bodlaender [Bo 92].

In [JS 1992] is proved that List Coloring is *NP*-complete when restricted to complete bipartite graphs. Hence it is *NP*-complete for comparability graphs of interval orders. But for graphs with fixed tree-width k there exists an algorithm whose running time is bounded by a polynomial of degree $k + 2$ in the number of vertices of the graph.

In the special case of comparability graphs of *series-parallel orders* Bodlaender and Möhring [BoM 93] showed that the tree-width can be determined by a linear time algorithm and that in this case tree-width equals path-width.

Our main result asserts that also the tree-width of comparability graphs of *interval orders* can be computed by a linear time algorithm and path-width equals tree-width for this class of graphs. “Linear” means linear in the size of the directed acyclic graph inducing the interval order.

In general path-width does not equal tree-width when we restrict to comparability graphs because we give an example of a comparability graph with different path-width and tree-width. This is in remarkable contrast with a result of Habib and Möhring [HM 92] who proved that in the case of *cocomparability graphs*, namely complements of comparability graphs, tree-width equals always path-width.

The result in [HM 92] has been extended to the class of *asteroidal triple free graphs* (cf. Möhring [M 93]). Comparability graphs of interval orders, however, are not necessarily asteroidal triple free.

Our algorithmic results improve a result of Kloks [Kl 93] who proves that the tree-width of *co-chordal graphs*, i.e., complements of chordal graphs, can be determined by an $O(n^2)$ and the path-width of co-chordal graphs by a polynomial algorithm.

2 Definitions and Notations

We treat (partial) orders induced by directed acyclic graphs. Let $G = (V, E)$ be a directed acyclic graph. The adjacency structure of G induces an order $P(G) = (V, <)$ on V by

$$x < y \quad \text{if} \quad (x, y) \in E.$$

Denote for each element $x \in V$ by $N(x) \stackrel{\text{def}}{=} \{y \in V \mid x < y\}$ its successor set and its predecessor set by $N^{-1}(x) \stackrel{\text{def}}{=} \{u \in V \mid u < x\}$.